1. 设复变函数 f 在区域 D 内有定义且实部虚部的一阶偏导数连续，$G \subset D$ 是其子区域并有 $G \cup \partial G \subset D$. 证明复变函数的格林公式. 即

$$\int_{\partial G} f(z,\bar{z})\,dz = \int_G \partial_{\bar{z}} f(z,\bar{z})\,d\bar{z}\,dz.$$

设 $f(z,\bar{z}) = u(x,y) + iv(x,y)$.

$$\int_{\partial G} f(z,\bar{z})\,dz = \int_{\partial G} u(x,y) + iv(x,y),\ d(x+iy)$$

$$= \int_{\partial G} (u\,dx - v\,dy) + i\int_{\partial G}(u\,dy + v\,dx) = -\int_{\partial G}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)dx\,dy + i\int_{\partial G}\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)dx\,dy$$

$$\partial_{\bar{z}} f = \frac{\partial f}{\partial(x-iy)} = \frac{\partial f\,\partial\bar{z}\,\partial z}{\partial z\,\partial\bar{z}}$$

$$f = f(z,\bar{z}) = f(x+iy,\ x-iy). = u(x,y) + iv(x,y)$$

$$df = \partial f$$